



**UNIVERSITÉ DE
FRANCHE-COMTÉ**

**ADVANCED DYNAMICAL SYSTEMS
MASTER 2 COMPUHYS**

Take home exam

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1 Properties of $\mathcal{H}_0$

$\mathcal{H}_0(\Theta)$ is the unperturbed part of the Hamiltonian and depends only on the momentum Θ . Being quadratic in Θ ($\mathcal{H}_0 = \frac{n\Theta^2}{2}$), it represents pure rotation and is completely integrable.

2 Symplectic change of variables

For the transformation $(\theta, \Theta) \rightarrow (\sigma, \Sigma)$ where $\sigma = \theta - \frac{i}{2}nt$ and $\Sigma = \Theta$:

$$d\theta \wedge d\Theta = d\sigma \wedge d\Sigma$$

since $d\sigma = d\theta$ (nt being a parameter, not a dynamical variable) and $d\Sigma = d\Theta$. Therefore, the transformation preserves the canonical form and is symplectic.

3 Nature of chaos

The system exhibits potential for chaos through resonance overlap. Each of the four resonances behaves like a pendulum, and when their separatrices overlap (Chirikov criterion), chaotic motion can emerge.

4 Location of resonances

For each resonance, we have the Hamiltonian:

$$K_i(\sigma, \Sigma) = n \frac{\Sigma^2}{2} - \frac{i}{2} n \Sigma - \frac{3n}{4} \frac{B-A}{C} f_i(e) \cos(2\sigma),$$

To locate the equilibria in (σ, Σ) -space, we need to solve:

$$\frac{\partial K_i}{\partial \sigma} = 0, \quad \text{and} \quad \frac{\partial K_i}{\partial \Sigma} = 0.$$

Firstly,

$$\frac{\partial K_i}{\partial \Sigma} = 0$$

$$n\Sigma - \left(\frac{i}{2}\right)n = 0$$

Therefore,

$$\Sigma_{\text{res}} = \frac{i}{2}$$

For σ , the derivative is:

$$\frac{\partial K_i}{\partial \sigma} = \frac{3n}{4} \frac{B-A}{C} f_i(e) \cdot 2 \sin(2\sigma).$$

Setting $\frac{\partial K_i}{\partial \sigma} = 0$, we find:

$$\sin(2\sigma) = 0 \implies 2\sigma = k\pi, \quad \sigma = \frac{k\pi}{2}, \quad k \in \mathbb{Z}.$$

The solutions within the interval $[0, 2\pi[$ are:

$$\sigma = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}.$$

To classify these equilibria, we compute the second derivative:

$$\frac{\partial^2 K_i}{\partial \sigma^2} = \frac{3n}{4} \frac{B-A}{C} f_i(e) \cdot 4 \cos(2\sigma).$$

- For $\cos(2\sigma) > 0$, the equilibrium is stable (libration center), here it happens at $\sigma = 0$ and $\sigma = \pi$.
- For $\cos(2\sigma) < 0$, the equilibrium is unstable (saddle point), here it happens at $\sigma = \frac{\pi}{2}$ and $\sigma = \frac{3\pi}{2}$,

Thus, for each resonance i , the libration centers occur at:

$$\begin{cases} \Sigma_{\text{res}} = \frac{i}{2} & (\text{resonant value}) \\ \sigma = 0, \pi & (\text{stable equilibria}) \\ \sigma = \frac{\pi}{2}, \frac{3\pi}{2} & (\text{unstable equilibria}) \end{cases}$$

5 Width of the resonances

For a resonance i , considering the pendulum-like Hamiltonian:

$$K_i(\sigma, \Sigma) = n \frac{\Sigma^2}{2} - \frac{i}{2} n \Sigma - \frac{3n}{4} \frac{B-A}{C} f_i(e) \cos(2\sigma),$$

The width $\Delta\Sigma$ can be derived by analyzing the separatrix:

1. At the unstable point ($\sigma = \frac{\pi}{2}$ or $\frac{3\pi}{2}$), the energy equals:

$$K_i(\sigma = \frac{\pi}{2}, \Sigma_{\text{res}} = \frac{i}{2}) = 0 + \frac{3n}{4} \left(\frac{B-A}{C} \right) f_i(e)$$

2. At maximum excursion (where $\sigma = 0$ or π), the energy must equal this same value since we're on the separatrix:

$$K_i(\sigma = 0, \Sigma_{\text{max}}) = \frac{n}{2} (\Sigma_{\text{max}}^2 - i \Sigma_{\text{max}}) - \frac{3n}{4} \left(\frac{B-A}{C} \right) f_i(e)$$

The energies must be equal,

$$\begin{aligned} K_i(\sigma = \frac{\pi}{2}, \Sigma_{\text{res}} = \frac{i}{2}) &= K_i(\sigma = 0, \Sigma_{\text{max}}) \\ \frac{3n}{4} \left(\frac{B-A}{C} \right) f_i(e) &= \frac{n}{2} (\Sigma_{\text{max}}^2 - i \Sigma_{\text{max}}) - \frac{3n}{4} \left(\frac{B-A}{C} \right) f_i(e) \\ \frac{n}{2} (\Sigma_{\text{max}}^2 - i \Sigma_{\text{max}}) &= \frac{3n}{2} \left(\frac{B-A}{C} \right) f_i(e) \\ \Sigma_{\text{max}}^2 - i \Sigma_{\text{max}} - 3 \left(\frac{B-A}{C} \right) f_i(e) &= 0 \\ \Sigma_{\text{max}} &= \frac{1}{2} \left(i \pm \sqrt{i^2 + 12 \left(\frac{B-A}{C} \right) f_i(e)} \right) \end{aligned}$$

Therefore, the width is:

$$\begin{aligned}
\Delta\Sigma &= \Sigma_{\max} - \Sigma_{\text{res}} \\
&= \frac{1}{2} \left(i \pm \sqrt{i^2 + 12 \left(\frac{B-A}{C} \right) f_i(e)} \right) - \frac{i}{2} \\
&= \pm \frac{1}{2} \sqrt{i^2 + 12 \left(\frac{B-A}{C} \right) f_i(e)}
\end{aligned}$$

Thus, we can estimate the width to be:

$$\Delta\Sigma_i = \frac{1}{2} \sqrt{i^2 + 12 \left(\frac{B-A}{C} \right) f_i(e)}$$

where, $i = 1, 2, 3, 4$, and $f_i(e)$ depends on the resonance:

$$\begin{aligned}
f_1(e) &= -\frac{e}{2}, & f_2(e) &= 1 - \frac{5e^2}{2}, \\
f_3(e) &= \frac{7e}{2}, & f_4(e) &= \frac{17e^2}{2}.
\end{aligned}$$

6 Numerical application

For the numerical application we use the following values for the eccentricity and $(B-A)/C$.

	Eccentricity	$(B-A)/C$
Mercury	0.21	9.4×10^{-5}
Mimas	0.02	0.06
Hyperion	0.1	0.8

Table 1: Constants values for Mercury, Mimas, and Hyperion

To observe the resonances, we plot in Figure 1 the potential $K_i(\sigma)$ for $i = 2$, using the parameters for Mimas, it shows a periodic structure typical of a pendulum-like system, with alternating stable and unstable points.

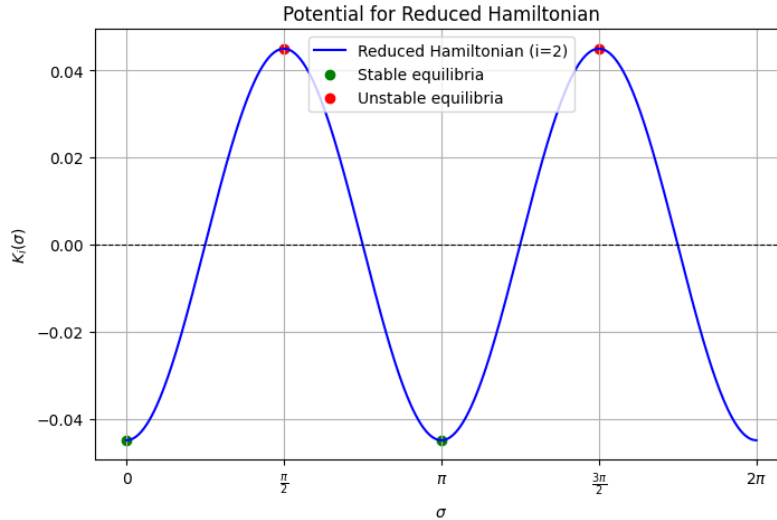


Figure 1: Potentiel for Mimas.

Now we compute the widths for each satellite, and the results are shown in Table 2 below.

	Resonance 1	Resonance 2	Resonance 3	Resonance 4
Mercury	0.5000	1.0001	1.5001	2.0000
Mimas	0.4982	1.0862	1.5042	2.0002
Hyperion	0.3606	1.8276	1.7578	2.0504

Table 2: Widths of Resonances in Σ -space

Mercury and Mimas show minimal overlap between resonances, as their widths remain close to the unperturbed values ($\approx 0.5, 1.0, 1.5, 2.0$). This is due to their small $(B - A)/C$ values (9.4×10^{-5} and 0.06 respectively). This allows them to maintain stable spin-orbit resonances: Mercury in 3:2 resonance and Mimas in synchronous rotation.

In contrast, **Hyperion's** resonances show significant deviations from these values and strong overlap, for all resonances 1, 2, 3, and 4. This extensive overlap, caused by its very large $(B - A)/C = 0.8$, and eccentricity $e = 0.1$, creates a wide chaotic zone preventing any stable spin-orbit configuration, including synchronous rotation.